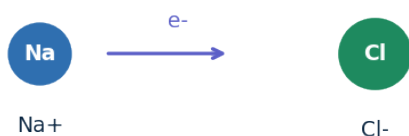


Chemical Bonding

JEE Chemistry — Notes, Formula Sheet, Practice Problems & Diagram

1. Notes

- Ionic bond forms by complete transfer of electrons between a metal and a non-metal, producing oppositely charged ions held by electrostatic attraction.
- Covalent bond forms by sharing of electron pairs between atoms (usually non-metals). Can be single, double, or triple bonds.
- VSEPR theory predicts molecular shape based on minimizing repulsion between electron pairs (bonding and lone pairs) around the central atom.
- Hybridization is the mixing of atomic orbitals to form new, equivalent hybrid orbitals: sp (linear), sp^2 (trigonal planar), sp^3 (tetrahedral).
- Molecular Orbital Theory: atomic orbitals combine to form bonding and antibonding molecular orbitals; bond order = $(\text{electrons in bonding MOs} - \text{electrons in antibonding MOs})/2$.
- Dipole moment measures bond/molecule polarity; it is zero for symmetric molecules (e.g., CO_2) even if individual bonds are polar.
- Hydrogen bonding is a special dipole-dipole attraction between H (bonded to N, O, or F) and a lone pair on another electronegative atom; responsible for water's high boiling point.



Electron transfer forms an ionic bond

Ionic bond formation: electron transfer from Na to Cl forms Na^+ and Cl^- .

2. Formula Sheet

Bond order (MOT)	$BO = (N_{\text{bonding}} - N_{\text{antibonding}}) / 2$
Dipole moment	$\mu = q \times d$
Formal charge	$FC = (\text{valence } e^-) - (\text{nonbonding } e^-) - (1/2)(\text{bonding } e^-)$
sp hybridization shape	Linear, 180 degrees
sp ² hybridization shape	Trigonal planar, 120 degrees
sp ³ hybridization shape	Tetrahedral, 109.5 degrees
Percentage ionic character (approx)	Higher electronegativity difference -> higher ionic character
Octet rule	Atoms tend to have 8 electrons in valence shell

3. Practice Problems (22 questions)

1. Predict the type of bond (ionic or covalent) formed between sodium and chlorine, with a reason.
2. Find the hybridization of the central atom in CH₄.
3. Find the hybridization of the central atom in BF₃.
4. Using VSEPR theory, predict the shape of NH₃ (with one lone pair on nitrogen).
5. Find the bond order of the N₂ molecule using molecular orbital theory (14 electrons total).
6. Explain why the dipole moment of CO₂ is zero even though the C=O bonds are polar.
7. Find the hybridization of carbon in C₂H₂ (acetylene, with a triple bond).
8. State the octet rule and give one example of a molecule that is an exception to it.
9. Explain why water (H₂O) has a bent shape rather than linear, using VSEPR theory.
10. Find the bond order of the O₂ molecule using molecular orbital theory (16 electrons total) and use it to explain O₂'s paramagnetism.
11. State the difference between a sigma bond and a pi bond.
12. Explain why hydrogen bonding in water leads to an unusually high boiling point compared to H₂S.
13. Find the number of lone pairs on the central atom in the water molecule.
14. Explain why BeCl₂ is linear while H₂O is bent, even though both have 2 bond pairs around the central atom.
15. Find the shape and hybridization of the central atom in PCl₅.
16. State Fajans' rules and explain how they predict increased covalent character in a bond.
17. Explain the difference between a polar covalent bond and a non-polar covalent bond, with one example of each.

18. Find the formal charge on the nitrogen atom in the ammonium ion (NH_4^+).
19. State the difference between bond order and bond length, and explain how they are related.
20. Explain why noble gases generally do not form chemical bonds.
21. Find the hybridization of sulfur in SF_6 .
22. Explain, using electronegativity difference, why HF is considered to have significant ionic character despite being a covalent molecule.

Answer Key

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| 1. Ionic - large electronegativity difference between Na (metal) and Cl (non-metal) favors electron transfer | 2. sp^3 |
| 3. sp^2 | 4. Trigonal pyramidal (due to one lone pair) |
| 5. Bond order = 3 (triple bond, very stable) | 6. Conceptual - CO_2 is linear and symmetric, so the two equal and opposite $\text{C}=\text{O}$ bond dipoles cancel |
| 7. sp | 8. Conceptual - e.g., BF_3 (incomplete octet) or SF_6 (expanded octet) are common exceptions |
| 9. Conceptual - 2 bond pairs + 2 lone pairs on O cause repulsion that bends the molecule to about 104.5 degrees | 10. Bond order = 2; O_2 has 2 unpaired electrons in antibonding π^* orbitals, explaining its paramagnetism |
| 11. Conceptual - sigma bond forms by head-on orbital overlap (single bond); pi bond forms by sideways overlap (present in double/triple bonds) | 12. Conceptual - O is more electronegative and smaller than S, so O-H hydrogen bonds are stronger than $\text{H}\cdots\text{S}$ interactions in H_2S |
| 13. 2 lone pairs | 14. Conceptual - BeCl_2 's central Be has no lone pairs (only 2 bond pairs, linear); H_2O 's O has 2 lone pairs, causing bending |
| 15. Trigonal bipyramidal, sp^3d hybridization | 16. Conceptual - Fajans' rules: small, highly charged cations and large, polarizable anions increase covalent character |
| 17. Conceptual - polar: unequal sharing, e.g., HCl; non-polar: equal sharing, e.g., Cl_2 | 18. Formal charge on N = +1 |
| 19. Conceptual - higher bond order generally means shorter, stronger bond length | 20. Conceptual - noble gases have a complete stable octet already, so there's little tendency to share or transfer electrons |
| 21. sp^3d^2 | 22. Conceptual - large electronegativity difference between H and F gives the H-F bond significant ionic character alongside covalent sharing |

Note: conceptual/derivation questions are marked accordingly in the answer key. Numerical answers are rounded for clarity — show full working in your own solutions.